

A Computationally Reproducible FC/dFC Workflow for Resting-State fMRI

Pilot Validation Using "Psychedelic rs-fMRI"

Dataset: PsiConnect / OpenNeuro ds006110

Current stage: workflow construction and pilot-run validation

127

Runs with matched
BOLD/confounds/mask files

1

Full pilot run processed

45

Windowed FC matrices
generated

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Current Progress: What Has Been Built?

Completed so far

- 1 Inspected fMRIPrep derivatives and built a resting-state file index.
- 2 Matched BOLD images, confounds files, and brain masks across runs.
- 3 Implemented metadata-based QC and readiness checks.
- 4 Validated ROI extraction, static FC, and sliding-window dFC on one pilot run.

Current contribution: a working and **computational reproducible** FC/dFC pipeline, not yet group-level inference.

Workflow overview

OpenNeuro / DataLad

BIDS + fMRIPrep

File Indexing



BOLD / Mask QC

ROI Parcellation

ROI Time-series



Static FC

Sliding-window dFC

Feature Extraction

Output of pilot validation

The workflow successfully generated

- 504×67 ROI time-series matrix
- 67×67 static FC matrix
- 45 sliding-window FC matrices

Why Dynamic Functional Connectivity?

Psychedelic-induced altered states may alter large-scale brain network organization.

Altered conscious experience unfolds over time; static FC may miss temporal dynamics.

Dynamic FC quantifies fluctuations in brain network coupling across time.

Research Questions

1. Can we build a documented and computationally reproducible FC/dFC workflow for processing this dataset?
2. After scaling up, can dFC features be compared across sessions or psychedelic-related conditions?

Static FC

One connectivity matrix per run
Captures average coupling
Useful baseline representation

Dynamic FC

Multiple matrices across time
Captures temporal fluctuations
Potentially relevant to altered consciousness

Dataset & Feasibility Inspection

Dataset

- **Dataset:** PsiConnect
- **OpenNeuro ID:** ds006110
- **Focus:** resting-state fMRI
- **Preprocessing:** fMRIPrep 22.0.2
- **Space:** MNI152NLin2009cAsym

Session coverage

65

ses-01 rest BOLD

62

ses-02 rest BOLD

Goal: determine whether the dataset contains matched derivatives for automated FC/dFC analysis.

FILE AVAILABILITY AFTER INDEXING AND PATH MATCHING

127

Resting-state MNI BOLD

127

Confounds TSV

127

Brain masks

Pilot Run QC and ROI Time Series

Pilot run

sub-PC001 / ses-01 / task-rest / run-1

The full workflow was first validated on one resting-state run before scaling up.

QC summary

BOLD shape **61 × 72 × 61 × 504**

Volumes **504** time points

TR **0.91 s**

Confounds rows **504**

Mean FD **0.114 mm**

Max FD **1.851 mm**

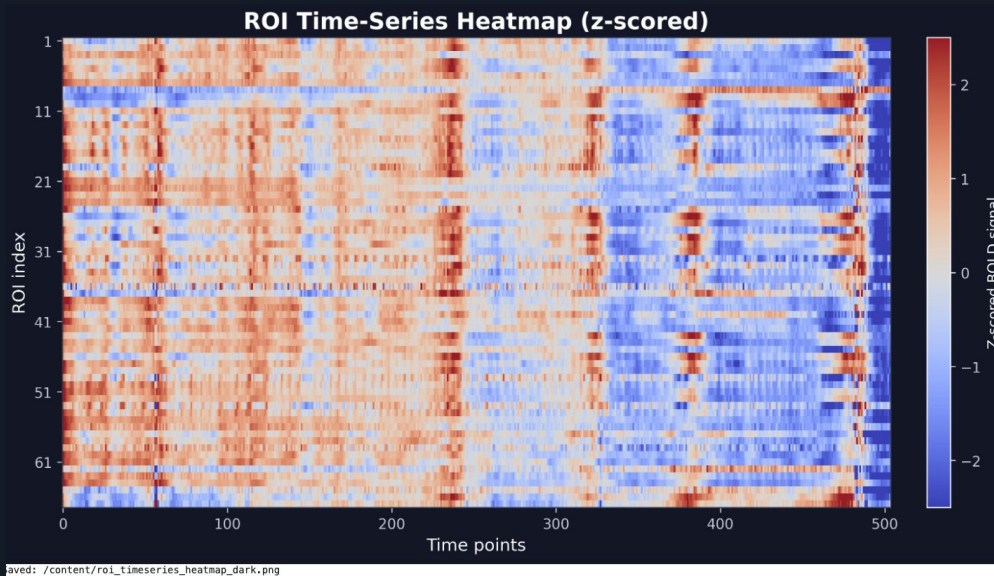
Ready for ROI **True**

ROI extraction

Preliminary grid atlas

ROI signals extracted and z-scored

Used for FC/dFC validation

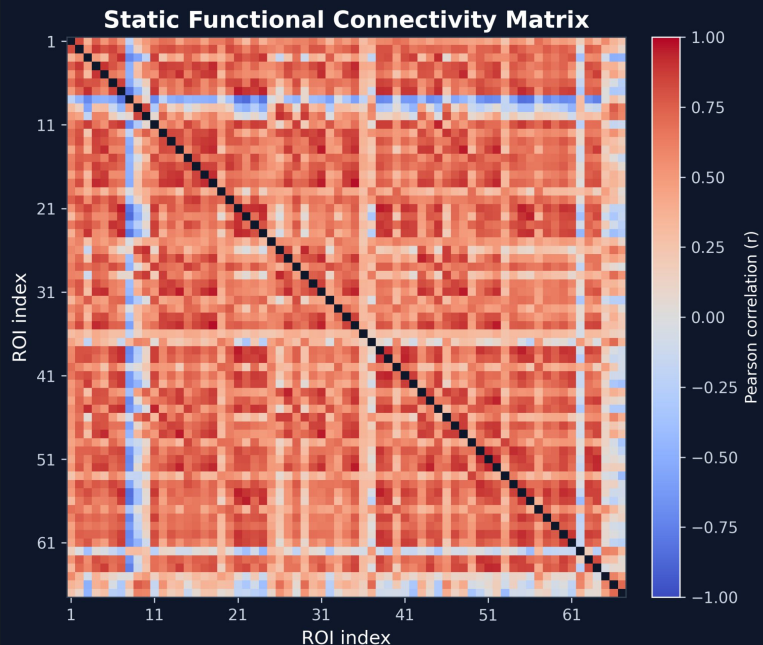


504 × 67 ROI time-series matrix

For final neurobiological interpretation, I plan to use a standard atlas such as Schaefer 100

Static Functional Connectivity

	ROI_001	ROI_002	ROI_003	ROI_004	ROI_005	ROI_006
ROI_001	1.0	0.8225211673424990	0.3727375565916310	0.755527886364819	0.5219398717173800	0.692679230594706
ROI_002	0.8225211673424990	1.0	0.5427374241046950	0.8877797224494440	0.7953669866884	0.8506910543219470
ROI_003	0.3727375565916310	0.5427374241046950	1.0	0.47122956757993700	0.31356277227522800	0.783525872500673



- Input: 504×67 ROI time-series matrix
- Method: pairwise Pearson correlation across the full run
- Output: 67×67 static FC matrix
- Diagonal masked for visualization
- Fisher z-transformed values will be used for statistical analysis

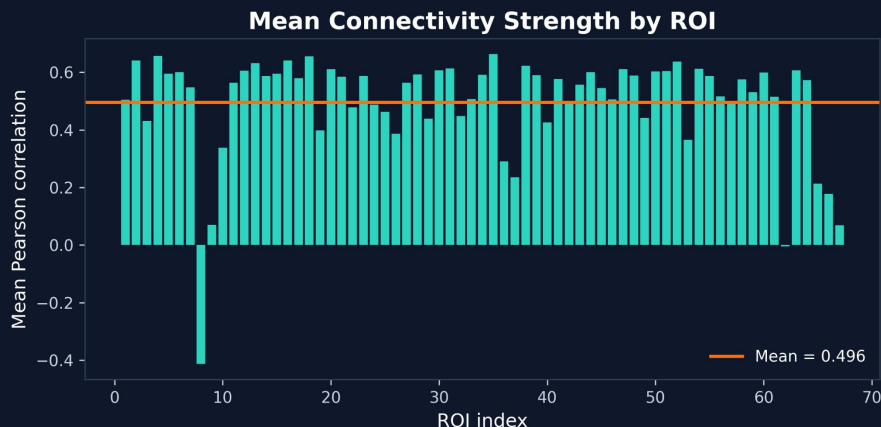
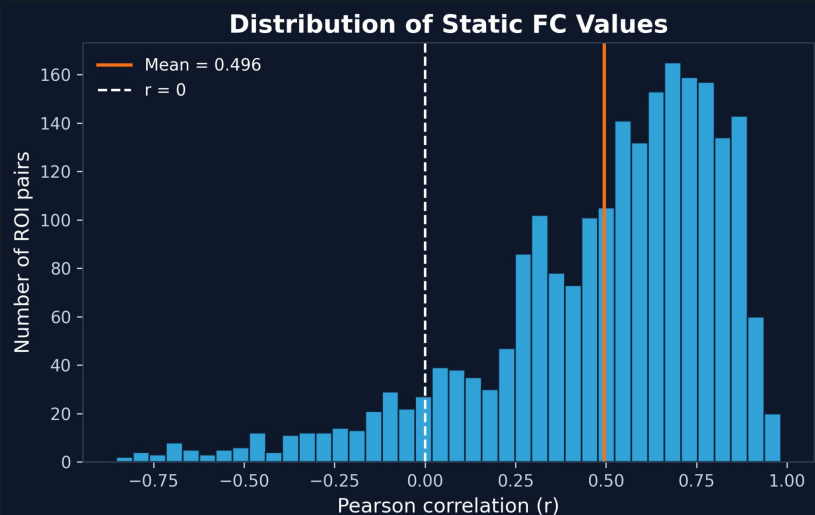
Interpretation

Static FC summarizes the average functional coupling across the entire resting-state run.

Used here as a static reference before estimating time-varying connectivity.

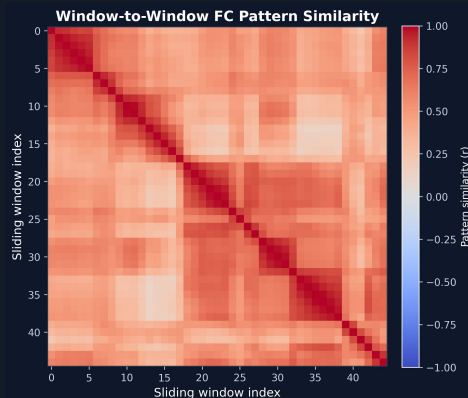
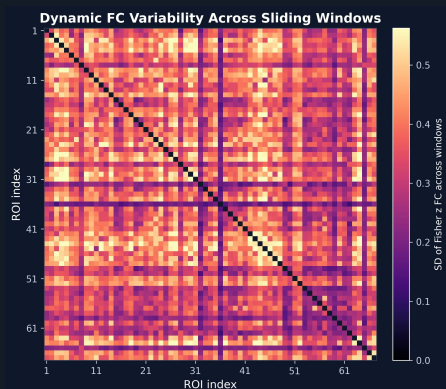
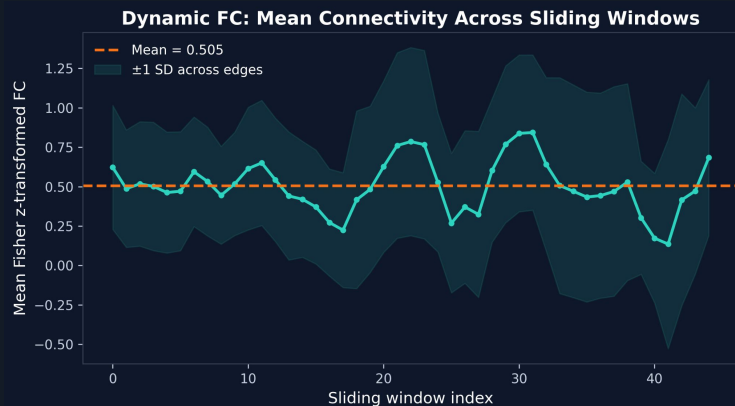
504×67 ROI time-series matrix $\rightarrow$ 67×67 Pearson correlation matrix

Static FC Descriptive Checks



The purpose here is mainly to check whether the static FC output looks reasonable, and whether the pipeline can generate basic edge-level and ROI-level summaries

Dynamic Functional Connectivity: Pilot Results



Parameters

Timepoints	504
Window size	60
Window duration	~54.6 s
Step size	10
Step duration	~9.1 s
Windows	45

Generated outputs

- 45 windowed FC matrices
- Window-level mean and SD of FC
- Edge-level dFC variability
- Window-to-window FC similarity

Mean edge variability
0.361

Max edge variability
0.653

Limitations & Next Steps

Current status

- One pilot run fully processed
- Dataset-level file matching completed
- Static FC and dFC outputs generated

Preliminary limitations

- Single-run validation only
- Preliminary grid-based atlas
- Motion/nuisance control not finalized
- No group-level inference yet
- Computational reproducibility only; not yet external replication

Next steps

- Scale to all 127 runs
- Use Schaefer 100/200 atlas
- Strengthen motion control
- Compare sessions/conditions

Take-home message:

A documented, computationally reproducible FC/dFC workflow has been pilot-validated and is ready for full-dataset processing.